Unit TimerPWR

SKU:U189



















Description

Unit TimerPWR is a timed power supply unit with key functions including charging and discharging, scheduled power switching, status display, and boosted output. Its built-in STM32 microcontroller handles the RTC (real-time clock) and overall control, allowing users to set automatic power-on and power-off times as needed.

The unit is powered through a **Type-C** interface and connects to an external rechargeable battery through a 1.25-2P connector. Its built-in battery charging circuit supports a charging current of 330mA. It also includes a DC-DC **boost** circuit that can supply 5V/800mA (when powered by a 1400mA@1C battery) to external devices through the Grove port. In addition, the built-in INA3221 sensor monitors input and output current and voltage in real time.

The unit features a 0.66-inch **OLED** display and two side buttons for user interaction, making it easy to view real-time system status and modify settings. Users can configure the unit with the side buttons or access the I2C bus through the Grove interface and use I2C commands to set power-on and power-off parameters. This product is suitable for smart homes, industrial automation, timed-control equipment, and similar applications.

Features

- RTC cyclic timer control
- Battery powered
- Built-in DC-DC boost circuit
- Current / voltage monitoring
- OLED display
- Grove interface
- Button-controlled power-on / power-off mode

Includes

- 1 x Unit TimerPWR
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

- Timed-control equipment
- Portable electronic devices
- Sensor node power supply

Specifications

Specification	Parameter
MCU	STM32G031G8U6
Input Voltage	Supports 5V DC input (through the Type-C interface)
Output Voltage	Built-in DC-DC boost circuit providing 5V output for external devices
Battery Power	3.7V lithium polymer battery
Display	0.66-inch OLED display, 64x48 resolution, SPI communication
Interfaces	Type-C interface: used to charge the battery (cannot directly power the Grove port) Grove interface: used for external sensor and module expansion, supporting communication and external power output Battery interface: 1.25mm@2P
I2C Communication Address	0x56
Current Sensor	INA3221
Charging Current	DC 5V@330mA
Charging IC	LGS4056H
Charging Temperature	55°C
Grove Port Max Output Current	Depends on battery capacity and discharge capability, e.g.: 1400mAh@1C 3.7V:DC4.96V@800mA 500mAh@1C 3.7V:DC4.97V@700mA 110mAh@1C 3.7V: DC5.03V@400mA
Standby Current	DC4.2V@36.67uA
Operating Temperature	0 ~ 40°C
Product Size	40.0 x 24.0 x 11.0mm
Product Weight	7.5g
Package Size	138.0 x 93.0 x 12.0mm
Gross Weight	13.2g

Learn

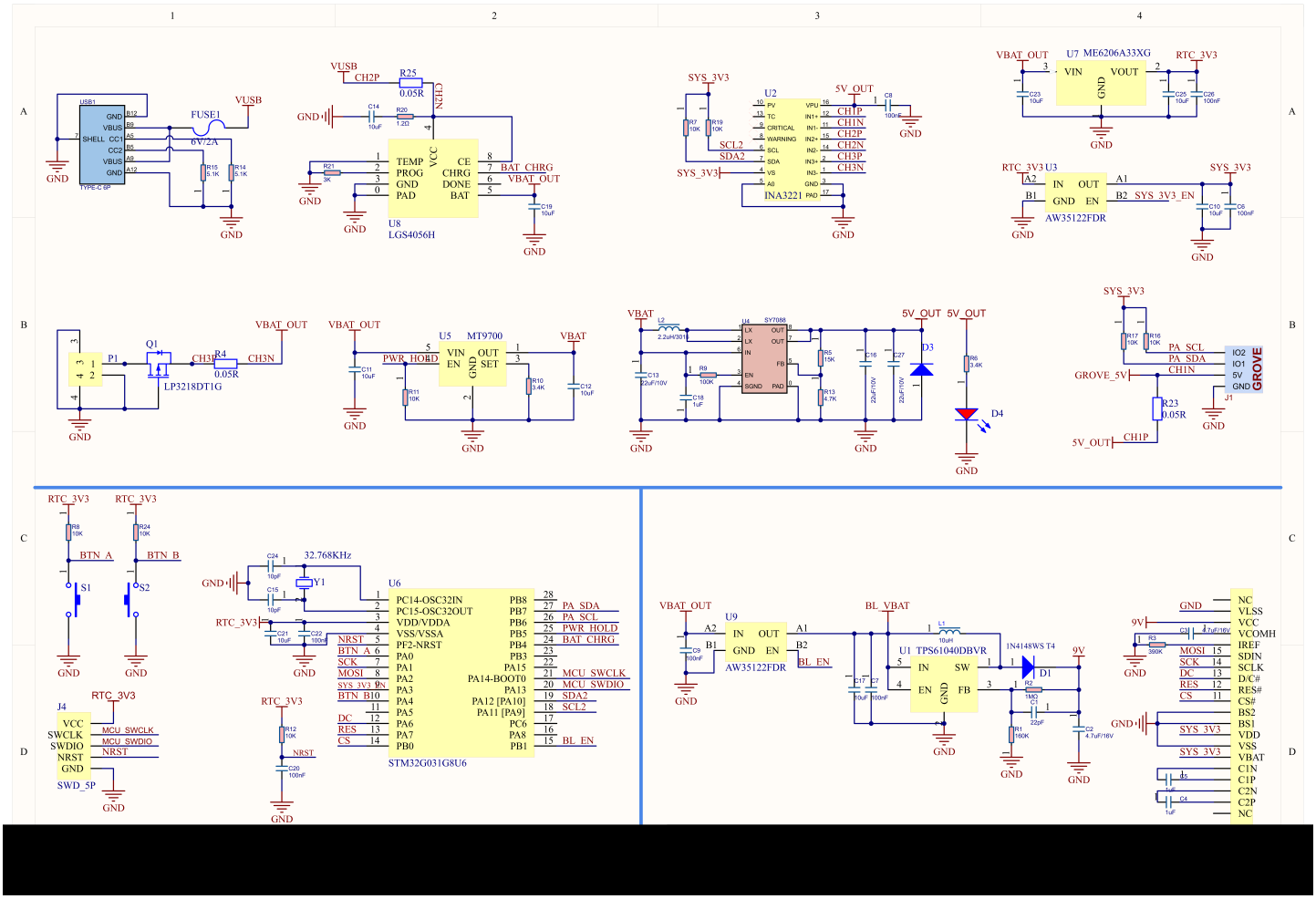
Power Supply

Power Supply Requirements

Unit TimerPWR must be connected to a lithium battery during use. Its USB interface is only used to charge the battery and cannot directly supply output power.

Schematics

- Unit TimerPWR Schematics PDF



PinMap

Unit TimerPWR

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

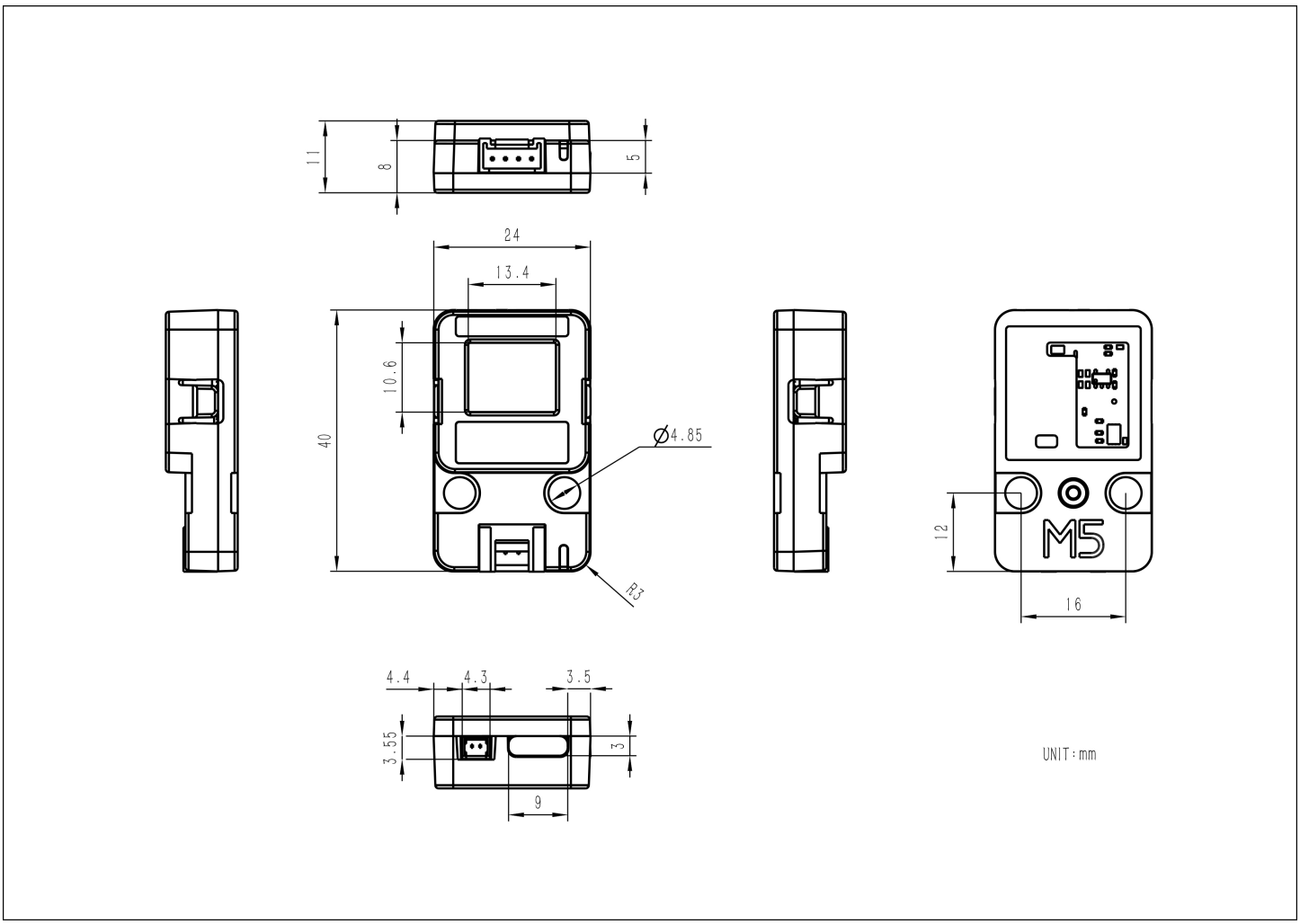
STM32G031G8U6

STM32G031G8U6	PB6	PB7	PA11	PA12	PB4	PA0	PA4
I2C1 (GROVE)	SCL1	SDA1					
I2C2 (INA3221)			SCL2	SDA2			
LED (Charging Indicator)					CHAG		
Button A						BTN_A	
Button B							BTN_B

OLED

STM32G031G8U6	PA2	PA1	PA6	PA7	PB0	PB1
OLED	OLED_MOSI	OLED_SCK	OLED_DC	OLED_RST	OLED_CS	BL_EN

Model Size



Structure

- Unit TimerPWR Structure Files

Datasheets

- INA3221
- LGS4056H

Softwares

Arduino

- Unit TimerPWR Arduino Library
- Unit TimerPWR Arduino Tutorial

UiFlow1

- Unit TimerPWR UiFlow1 Docs

UiFlow2

- Unit TimerPWR UiFlow2 Docs

Internal Firmware

- Unit TimerPWR Internal Firmware

Protocol

- Unit TimerPWR I2C Protocol PDF

M5Stack TimerPWR I2C Protocol																		V1 (FW Version) 2024/3/11
REG MAP (0x56)			0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Control	Sleep Command	0x40 W	Sleep Command																write 1, sleep
	Cycle	0x40 W/R		Cycle															1: Start Cycle 0: Stop Cycle
Config	Power control	0x00 W/R	Grove 5V Out Enable	OLED Backlight Enable															Grove 5V Out Enable: 0: Grove 5V out off 1: Grove 5V out on OLED Backlight Enable: 0: BL_OFF 1: BL_ON
	Wake&Sleep trigger	0x20 W/R	Wake Trigger	Sleep Trigger															Wake Trigger: bit 0: 1, TIMER_TRIGGER Enable; 0, TIMER_TRIGGER Disable; bit 1: 1, BUTTON_TRIGGER Enable; 0, BUTTON_TRIGGER Disable; Sleep Trigger: bit 0: 1, TIMER_TRIGGER Enable; 0, TIMER_TRIGGER Disable; bit 1: 1, BUTTON_TRIGGER Enable; 0, BUTTON_TRIGGER
	Power on Time	0x30 W/R	Hour	Min	Sec														
	Power off Time	0x80 W/R	Hour	Min	Sec														
Read	btn	0x10 R	btnA	btnB															0: press; 1: no press
	Voltage&Current	0x60 R	USB Voltage-byte0	USB Voltage-byte1	USB Voltage-byte2	USB Voltage-byte3	USB Current-byte0	USB Current-byte1	USB Current-byte2	USB Current-byte3	Grove Voltage-byte0	Grove Voltage-byte1	Grove Voltag-byte2	Grove Voltage-byte3	Grove Current-byte0	Grove Current-byte1	Grove Current-	Grove Current-byte3	Voltage&Current = Voltage&Current-byte0 + Voltage&Current-byte1 * 256 + Voltage&Current-byte2 * 65536 + Voltage&Current-byte3 * 16777216
	Voltage&Current	0x70 R	Battery Voltage-byte0	Battery Voltage-byte1	Battery Voltage-byte2	Battery Voltage-byte3	Battery Current-byte0	Battery Current-byte1	Battery Current-	Battery Current-byte3									The unit of voltage is V. The unit of current is mA. The true value needs to be divided by 100.
	Battery Charge Status	0x90 R	Battery Charge Status																0: no charge 1: charge
System	Flash Write back	0xF0 R	write back																write 1, write back
	Firmware Version	0xF0 R															FW Version		Value: firmware version
	I2C Addr	0xF0 R																I2C Addr	Value: I2C Address

Video

- Unit TimerPWR Product Introduction and Example Demonstration

timerpwr video.mp4

- UiFlow2 Use